

PRIYARANJAN PATTNAYAK

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| Website: <https://priyanpattnayak.com> | Google Scholar: <https://scholar.google.com/citations?user=gXWbDuUAAAAJ>

AI researcher and technology leader specializing in generative AI, multilingual (Indic & South Asian) NLP, and large-scale AI systems with 13+ years of experience. Currently Senior Principal Scientist at Oracle Cloud Infrastructure leading enterprise-scale AI initiatives in conversational AI, multilingual large language models, and AI observability platforms.

Author of **21+ peer-reviewed publications** and **14+ patents filed** in areas including multilingual NLP, generative AI architectures, and AI robustness. Active contributor to the global AI research community as **Area Chair for the ACL Industry Track**, reviewer for top-tier conferences including **ACL, NeurIPS, WWW, KDD, and CVPR**, and mentor in the **AAAI Undergraduate Consortium**.

Research interests: Generative AI systems and RAG architectures, Multilingual NLP and low-resource language modeling, AI safety and robustness evaluation for LLMs & Enterprise conversational AI platforms

PROFESSIONAL EXPERIENCE

Oracle Cloud Infrastructure, Seattle, USA

Jul 2018 – Present

Senior Principal Data Scientist ~ OCI AI

Generative AI & NLP Systems

- Architected **enterprise generative AI solutions using RAG and chain-of-thought reasoning**, enabling automated technical support and operational intelligence across Oracle Cloud services.
- Designed hybrid conversational AI architectures combining **LLMs, retrieval systems, and dynamic routing frameworks** to support multi-turn dialogue systems.
- Developed multilingual NLP pipelines and tokenization strategies to improve performance of language models across diverse languages.

AI Platforms & Large-Scale ML Systems

- Built a **managed ML platform** for Oracle Cloud TechOps enabling automated issue detection, classification, and resolution.
- Developed NLP pipelines using **topic modeling, clustering, and semantic similarity models** to analyze large-scale support ticket data and identify systemic issues.
- Designed time-series forecasting models for operational workload prediction and resource planning.

Leadership & Organizational Impact

- Lead AI initiatives across a VP organization of **500+ engineers** and guided ML strategy for enterprise automation systems.
- Managed and mentored **15+ data scientists and engineers** delivering production AI systems.
- Oracle **Bar-Raiser for technical hiring**, conducting **100+ ML and data science interviews**.

Impact: AI systems developed by the team are projected to generate **\$100M+ annual operational value**.

University of Washington CoMotion, Seattle, USA

2017 – 2018

Data Science Manager, Innovation Fund

- Supported UW innovation and commercialization work by translating technical ideas into user discovery, market assessment, and data-informed product strategy artifacts.
- Contributed to startup and incubation-oriented project work across analytics, product framing, and technology commercialization within the UW innovation ecosystem.

SEAL Lab, University of Washington, Seattle, USA

2017 – 2018

Graduate Researcher

- Conducted applied data science research in a university lab setting, building experimental analysis workflows and communicating findings to academic and industry stakeholders.
- Developed predictive modeling and evaluation artifacts for early applied ML work, including Pacific Northwest rainfall analysis and time-series forecasting research.

Data Science (@ T-Mobile)

- Designed a RNN based approach to detect contextual network anomaly in T-Mobile's US coverage.
- Mentored summer interns, defined scope of their work & headcount allocation to model development/maintenance.

Tata Consultancy Services (TCS), Bangalore, India

Lead

Information Management (*Client – American Express*)

Sep 2013 – Aug 2017

- Led a team to deploy time series forecasting model in R for capacity planning of SQL Server, Azure SQL & MongoDB.
- Managed re-engineering initiative to minimize SAN and resource utilization in SQL Server databases and data warehouses, which ended up saving 85000\$ for IM in 2015; Created dimensional data models for data warehouse systems.
- Awarded with TATA Champion Badge for creating a body of knowledge for MongoDB; helped the Centre of Excellence (CoE) in setting up RA (Reference Architecture) for MongoDB.

RESEARCH & PUBLICATIONS

Author of **21+ peer-reviewed research papers** in top-tier AI/NLP conferences including ACL, EMNLP, NAACL, EACL, AACL, and IEEE.

Selected research contributions include:

- **IndicJR: Benchmarking Jailbreak Robustness in South Asian Languages** – Introduced a multilingual benchmark covering multiple Indic languages to evaluate safety vulnerabilities in large language models. EACL 2026
- **Tokenization Matters: Improving Zero-Shot NER for Indic Languages** – Demonstrated the impact of tokenization strategies on generalization performance for low-resource languages. IEEE EIT 2025
- **Hybrid AI Architectures for Multi-Turn Conversational Systems** – Proposed enterprise conversational frameworks integrating LLM reasoning with retrieval pipelines. NAACL 2025
- **Large Multimodal Model Dataset Survey** – Comprehensive analysis of datasets and architectures used for multimodal AI systems. Springer Nature 2025
- **Mvtamperbench**: Evaluating robustness of vision-language models. ACL 2025
- **SweEval**: Do LLMs Really Swear? NAACL 2025
- **Crowdsource, Crawl, or Generate? Creating SEA-VL**: Multicultural Vision Language Dataset for Southeast Asia. ACL 2025

Research focuses on improving **robustness, multilingual capabilities, and real-world deployment of large language models**.

Google Scholar profile includes full list of publications spanning NLP, machine learning systems, and AI applications.

Key Research Contributions

- Multilingual NLP for low-resource **Indic** and **South Asian** languages, including benchmarks and tokenization strategies improving LLM performance. Safety research directly incorporated into a leading AI product for Indic languages.
- Developed **IndicJR**, the first jailbreak robustness benchmark for large language models across multiple Indic languages, highlighting safety vulnerabilities in multilingual AI systems.
- Proposed tokenization strategies enabling **zero-shot NER for extremely low-resource languages such as Santhali**, improving LLM generalization in multilingual settings.
- Enterprise generative AI architectures combining retrieval-augmented generation (RAG), chain-of-thought reasoning, and dynamic routing. Production deployment of conversational AI and ML systems for large-scale cloud infrastructure operations.

PATENTS & INNOVATION

Inventor on **14+ patents filed** (and counting) related to AI systems and generative AI architectures.

Areas of innovation include:

- Conversational AI infrastructure
- AI observability and monitoring frameworks
- Large language model orchestration systems
- Automated support resolution systems

These innovations focus on **operationalizing generative AI in enterprise environments at scale**.

PROFESSIONAL SERVICE & COMMUNITY LEADERSHIP

Conference Leadership

Area Chair — ACL Industry Track 2026

Program Committee / Reviewer for A* AI/NLP conferences: ACL ARR, CVPR, KDD, WWW, NeurIPS etc.

RL-Eval Workshop co-organizer at ACM CAIS 2026.

Mentorship:

AAAI Undergraduate Consortium mentor for NSF Grants

Mentor undergrad and graduate student researchers across universities in US and India

INVITED TALKS, ACADEMIC ENGAGEMENT & MEDIA ARTICLES

Frequent speaker at academic and industry venues on generative AI systems and NLP research.

Topics: Enterprise Generative AI Systems, Multilingual (**Indic**) NLP and Low-Resource Language Modeling, Responsible AI

Speaker/ Talks:

- Why Agents Fail? - [Conf42 Machine Learning 2026](#)
- Future of Gen AI - [Tech & AI Conference NYC 2025](#)
- Guest Lecture - University of Washington Seattle on Cloud & AI, Fall 2025
- Generative AU at Oracle Cloud - [Everything Product Podcast](#)
- Enterprise Chatbots & Hybrid AI - [NAACL 2025 Oral Presentation](#)
- Industry Panel at 28th Colloquium for Information Systems Security - University of Tampa, FL, 2024

Media Articles:

- LLM safety for Indic and Low resource languages: [Gurgaon Mail 27-Oct-2025 News print](#)
- From Odisha to Seattle: The Indian Scientist Teaching AI how Indians Really Talk: [Indian Express Print 01-Feb-2026](#)

EDUCATION

University of Washington, Seattle WA

Master of Science in Information Systems (Data Science) – Graduated w/ Honors (Top 5%)

Jun 2018

(GPA: 3.84/4.00)

Biju Patnaik University of Technology - College of Engineering and Technology

Bachelor of Technology – Electrical Engineering

Jul 2013

(GPA: 3.95/4.00)